DR B R AMBEDKAR NATIONAL INSTITUTE OF TECHNOLOGY JALANDHAR

Department of Computer Science and Engineering

SUMMER INTERNSHIP REPORT

on

HappyClassify & HappyPredict: Machine Learning-Based Classification and Regression Analysis of World Happiness Levels Using Socio-Economic Indicators

Submitted as a record of the work carried out during the
Summer Internship Programme at NIT Jalandhar

Submitted by

Shaurya Rana

Roll Number: 2510104065
Bachelor of Technology in Computer Science and Engineering (Artificial Intelligence & Machine Learning)
First Year, Second Semester
Department of Computer Science and Engineering(AIML)
Jawaharlal Nehru Government Engineering College
Sundar Nagar, Himachal Pradesh

Under the Guidance of

Dr. Rajneesh Rani

Associate Professor
Department of Computer Science and Engineering
Dr B R Ambedkar National Institute of Technology Jalandhar

Internship Duration

Four Weeks
22 June 2026 to 22 July 2026

July 2026

Certificate

This is to certify that **Shaurya Rana**, a student of **Bachelor of Technology in Computer Science and Engineering (Artificial Intelligence & Machine Learning)** at **Jawaharlal Nehru Government Engineering College**, has successfully completed the Summer Internship Programme at the **Department of Computer Science and Engineering, Dr B R Ambedkar National Institute of Technology Jalandhar**, during the period from **22 June 2026** to **22 July 2026**.

During the internship, the student worked on the project entitled

"HappyClassify & HappyPredict: Machine Learning-Based Classification and Regression Analysis of World Happiness Levels Using Socio-Economic Indicators"

under my guidance.

The student was involved in understanding the assigned problem, studying relevant concepts and research literature, dataset preparation, implementation, experimentation, analysis of results, and preparation of the internship report.

To the best of my knowledge, the work presented in this report is based on the activities carried out by the student during the internship period and is satisfactory for submission as a Summer Internship Report.

Dr. Rajneesh Rani
Associate Professor
Department of Computer Science and Engineering
Dr B R Ambedkar National Institute of Technology Jalandhar

Date: ____________________

Place: NIT Jalandhar

Declaration

I, **Shaurya Rana**, hereby declare that the Summer Internship Report entitled **"HappyClassify & HappyPredict: Machine Learning-Based Classification and Regression Analysis of World Happiness Levels Using Socio-Economic Indicators"** is based on the work carried out by me at the **Department of Computer Science and Engineering, Dr B R Ambedkar National Institute of Technology Jalandhar**, under the guidance of **Dr. Rajneesh Rani**, during the period from **22 June 2026** to **22 July 2026**.

I further declare that:

1. The work presented in this report was completed by me during the Summer Internship Programme.
2. This report has not been submitted, either in full or in part, to any other institution for the award of any degree, diploma, academic credit, or certificate.
3. All datasets, software tools, research papers, books, websites, source-code repositories, and other resources used in this work have been appropriately acknowledged.
4. The experimental results presented in this report have not been fabricated, manipulated, or misrepresented.
5. Any use of generative artificial intelligence tools has been disclosed in the relevant section of this report.
6. I remain responsible for the correctness, originality, and integrity of the work presented in this report.

Shaurya Rana
Roll Number: 2510104065
Jawaharlal Nehru Government Engineering College

Date: ____________________

Place: _________________________

Acknowledgement

I express my sincere gratitude to **Dr. Rajneesh Rani**, **Associate Professor**, Department of Computer Science and Engineering, Dr B R Ambedkar National Institute of Technology Jalandhar, for providing me with the opportunity to undertake my Summer Internship under her guidance.

Her valuable suggestions, technical guidance, encouragement, and constructive feedback helped me to understand the assigned problem and complete the internship work systematically.

I would also like to sincerely thank **Mr. Digvijay Singh** and **Miss Farhat Mir**, M.Tech. research scholars in the Department of Computer Science and Engineering, NIT Jalandhar, for their continuous technical support, patient guidance, and helpful discussions throughout the internship, which greatly contributed to the successful completion of this project.

I am also thankful to the faculty members, research scholars, laboratory staff, and fellow interns of the Department of Computer Science and Engineering, NIT Jalandhar, for their support and cooperation during the internship.

I acknowledge **Jawaharlal Nehru Government Engineering College** for permitting and encouraging me to undertake this Summer Internship.

Finally, I thank my family and friends for their continuous support and encouragement.

Shaurya Rana

Abstract

This report presents the work carried out during a four-week Summer Internship on the development of **HappyClassify** and **HappyPredict**, two complementary machine-learning applications built on the World Happiness Report (WHR) dataset. The internship addressed the problem of estimating national well-being from socio-economic indicators through two formulations: a multi-class classification task that assigns each country to one of four happiness levels (Unhappy, Moderately Happy, Happy, and Very Happy), and a regression task that predicts the continuous Life Ladder happiness score directly. The dataset, drawn from the latest edition of the World Happiness Report, covers approximately 143 countries and nine socio-economic features, including GDP per capita, social support, healthy life expectancy, freedom to make life choices, generosity, perceptions of corruption, positive and negative affect, and the Dystopia residual. The data was cleaned, missing values were imputed, outliers were treated using the interquartile-range method, and features were standardised before being split into stratified training, validation, and testing sets. Six classification algorithms (Logistic Regression, K-Nearest Neighbors, Decision Tree, Random Forest, Support Vector Machine, and XGBoost) and four regression algorithms (Linear Regression, Decision Tree Regressor, Random Forest Regressor, and Support Vector Regressor) were implemented and compared. The Random Forest algorithm produced the strongest results on both tasks, correctly classifying happiness levels with an accuracy of approximately 83 percent on the held-out test set. Both models were deployed as interactive Streamlit dashboards offering exploratory data analysis, model comparison, diagnostic visualisations, and real-time prediction from user-adjustable feature sliders. This report documents the background, dataset, methodology, implementation, experimental setup, results, and learning outcomes of the internship, along with the engineering challenges encountered during development and deployment.

Keywords: Summer Internship, Machine Learning, World Happiness Report, Classification, Regression, Random Forest, Streamlit, Exploratory Data Analysis

Contents

List of Figures

List of Tables

List of Abbreviations

Abbreviation	Meaning
AI	Artificial Intelligence
ML	Machine Learning
WHR	World Happiness Report
EDA	Exploratory Data Analysis
GDP	Gross Domestic Product
KNN	K-Nearest Neighbors
SVM	Support Vector Machine
SVR	Support Vector Regressor
IQR	Interquartile Range
CSV	Comma-Separated Values
MAE	Mean Absolute Error
MSE	Mean Squared Error
RMSE	Root Mean Squared Error
R^2	Coefficient of Determination
ROC	Receiver Operating Characteristic
AUC	Area Under the Curve
TP	True Positive
TN	True Negative
FP	False Positive
FN	False Negative
GUI	Graphical User Interface

Chapter 1

Introduction

1.1 Background

Subjective well-being, commonly measured through self-reported life-satisfaction scores, has become an important complement to purely economic indicators such as Gross Domestic Product when evaluating national progress and the effectiveness of public policy. The World Happiness Report, published annually by the Sustainable Development Solutions Network, quantifies this well-being through the Cantril Life Ladder score, a scale from 0 to 10 on which survey respondents rate their current lives, and pairs it with socio-economic indicators such as GDP per capita, social support, healthy life expectancy, freedom to make life choices, generosity, and perceptions of corruption.

From a machine-learning perspective, the relationship between these indicators and reported happiness can be studied in two complementary ways. It can be treated as a **classification** problem, where countries are grouped into discrete, policy-interpretable happiness tiers, or as a **regression** problem, where the exact happiness score is predicted as a continuous value. This internship explored both formulations using supervised machine-learning techniques and packaged the resulting models into interactive web dashboards.

1.2 Problem Statement

Given a set of socio-economic indicators for a country – namely GDP per capita, social support, healthy life expectancy, freedom to make life choices, generosity, perceptions of corruption, positive affect, negative affect, and the Dystopia residual – the internship addressed two related prediction problems:

- **Classification (HappyClassify):** predict the happiness level of a country, categorised into four ordinal classes – Unhappy, Moderately Happy, Happy, and Very Happy – based on its Life Ladder score.
- **Regression (HappyPredict):** predict the exact, continuous Life Ladder happiness

score of a country.

The problem is important because it demonstrates how publicly available socio-economic data can be used to build interpretable, data-driven tools for comparative well-being analysis. The primary technical challenges involved handling a relatively small, survey-based dataset, avoiding overfitting given the limited number of countries, and building a system that could serve predictions interactively rather than as a one-off script.

1.3 Motivation

Understanding which socio-economic factors most strongly influence national happiness is valuable for policy makers, researchers, and the general public. Most existing academic treatments of the World Happiness Report data use either purely statistical regression analysis or a single machine-learning model, and are rarely presented as an interactive tool. This internship was motivated by the opportunity to build a complete, end-to-end system – from raw data to a deployed interactive dashboard – that lets a user explore the data, compare multiple models, and generate live predictions for hypothetical or real socio-economic profiles.

1.4 Objectives

The main objectives of the summer internship are:

1. To understand the fundamental concepts related to classification and regression in machine learning.
2. To perform data cleaning and preprocessing on the World Happiness Report dataset.
3. To conduct exploratory data analysis (EDA) of the dataset.
4. To engineer meaningful features, including the happiness level target used for classification.
5. To build and compare multiple classification and regression models.
6. To evaluate the models using appropriate performance metrics.
7. To identify the key socio-economic factors influencing happiness levels.
8. To develop an interactive prediction dashboard and prepare a conference-style research paper documenting the work.

1.5 Scope of the Internship

The internship covered the complete pipeline from raw data to a deployed application:

- **Tasks completed:** dataset acquisition and cleaning, exploratory data analysis, feature engineering, training and comparison of six classification models and four regression models, model evaluation and diagnostic visualisation, and development and deployment of two interactive Streamlit dashboards.
- **Dataset and models considered:** the World Happiness Report (latest edition), Logistic Regression, K-Nearest Neighbors, Decision Tree, Random Forest, Support Vector Machine, and XGBoost for classification; Linear Regression, Decision Tree Regressor, Random Forest Regressor, and Support Vector Regressor for regression.
- **Software tools used:** Python, pandas, NumPy, scikit-learn, XGBoost, Matplotlib, Seaborn, Plotly, Streamlit, and Jupyter/JupyterLab under the Anaconda distribution.
- **Outside the scope:** deep-learning approaches, external cross-dataset validation, and large-scale hyperparameter search were outside the scope of this four-week internship.

1.6 Organisation of the Report

This report is organised as follows. Chapter 1 introduces the problem, motivation, objectives, and scope of the internship. Chapter 2 reviews the relevant literature and technical concepts. Chapter 3 describes the dataset and preprocessing steps. Chapter 4 presents the methodology and implementation. Chapter 5 explains the experimental setup. Chapter 6 presents and discusses the results. Chapter 7 summarises the work, learning outcomes, limitations, and future directions. The final chapter records the weekly progress of the internship.

Chapter 2

Literature Review and Technical Background

2.1 Technical Concepts

This section summarises the concepts required to understand the internship work.

- **Supervised learning:** learning a mapping from input features to a known output (label or value) using labelled training data.
- **Classification:** predicting a discrete category, such as one of the four happiness levels used in HappyClassify.
- **Regression:** predicting a continuous numeric value, such as the Life Ladder score predicted by HappyPredict.
- **Ensemble methods:** combining multiple weak learners (for example, decision trees) into a stronger model, as done by Random Forest and XGBoost.
- **Feature scaling:** transforming numeric features onto a common scale so that no single feature dominates model training purely due to its magnitude.
- **Train-validation-test split:** partitioning data so that model selection and final evaluation are performed on data the model has not seen during training.
- **Performance evaluation:** quantifying model quality using metrics such as accuracy and F1-score for classification, and mean absolute error and R^2 for regression.

2.2 Review of Existing Work

Prior work on the World Happiness Report data broadly falls into two categories. Statistical and econometric studies, largely originating from the WHR's own authors and affiliated researchers, use ordinary least squares regression and panel-data models to attribute variation in the Life Ladder score to specific socio-economic predictors, and consistently identify GDP per capita, social support, and healthy life expectancy as the strongest predictors [**?**].

Separately, machine-learning-oriented studies have applied ensemble methods such as Random Forest and Gradient Boosting to the same or similar feature sets, generally finding that ensemble tree-based methods outperform linear models on this data because of non-linear interactions between predictors, for example the diminishing marginal effect of GDP per capita on happiness at high income levels [**?**]. Classification-based framings of the problem, where countries are grouped into discrete happiness tiers, are comparatively less common in the literature.

Table 2.1: Comparison of existing approaches to World Happiness Report modelling

Approach	**Dataset**	**Method**	**Typical Finding**	**Limitation**
Statistical / econometric studies	WHR panel data	OLS regression, fixed-effects models	GDP, social support, and life expectancy are top predictors	Assumes linear relationships
Machine-learning regression studies	WHR panel data	Random Forest, Gradient Boosting	Ensemble methods outperform linear regression	Rarely deployed as an interactive tool
Machine-learning classification studies	WHR-derived happiness tiers	Various classifiers	Less commonly studied than regression	Few comparative, multi-model evaluations

2.3 Research or Implementation Gap

Based on this review, most prior work addresses either the classification or the regression formulation of the problem in isolation, and is typically presented as a static analysis rather than a usable tool. The gap addressed during this internship is the joint treatment of both formulations – HappyClassify and HappyPredict – within a single, deployed, interactive system that allows a non-technical user to explore the data, compare multiple models side by side, and obtain a live prediction from adjustable inputs.

Chapter 3

Dataset and Data Preparation

3.1 Dataset Description

- **Name of the dataset:** World Happiness Report (latest edition).
- **Source:** Sustainable Development Solutions Network, World Happiness Report official release.
- **Number of samples:** approximately 143 countries.
- **Number of classes (classification task):** four ordinal happiness levels.
- **Input data type and dimensions:** nine numeric socio-economic features per country.
- **Class distribution:** moderately imbalanced, with fewer countries in the "Unhappy" and "Very Happy" extremes compared to the two middle classes.
- **Dataset licence:** publicly released by the World Happiness Report for research and educational use.
- **Ethical and privacy considerations:** the dataset contains only country-level aggregate statistics and no personally identifiable information.

The nine input features used for both the classification and regression models are: GDP per capita, Social support, Healthy life expectancy, Freedom to make life choices, Generosity, Perceptions of corruption, Positive affect, Negative affect, and Dystopia residual. For the classification task, the continuous Life Ladder score was converted into four ordinal classes as shown in Table 3.1.

Table 3.1: Happiness level classification thresholds

Class	Label	Life Ladder Score Range
0	Unhappy	score < 5
1	Moderately Happy	5 ≤ score < 6
2	Happy	6 ≤ score < 7
3	Very Happy	score ≥ 7

3.2 Sample Data

	Country name	year	Life Ladder	Log GDP per capita	Social support	Healthy life expectancy at birth	Freedom to make life choices	Generosity	Perceptions of corruption
19	Albania	2011	5.867	9.31	0.759	67.88	0.487	-0.209	0.877
20	Albania	2012	5.51	9.326	0.785	68.16	0.602	-0.173	0.848
21	Albania	2013	4.551	9.338	0.759	68.44	0.632	-0.131	0.863
22	Albania	2014	4.814	9.358	0.626	68.72	0.735	-0.029	0.883
23	Albania	2015	4.607	9.382	0.639	69.0	0.704	-0.085	0.885
24	Albania	2016	4.511	9.417	0.638	69.025	0.73	-0.021	0.901
25	Albania	2017	4.64	9.455	0.638	69.05	0.75	-0.033	0.876
26	Albania	2018	5.004	9.497	0.684	69.075	0.824	0.005	0.899
27	Albania	2019	4.995	9.522	0.686	69.1	0.777	-0.103	0.914
28	Albania	2020	5.365	9.494	0.71	69.125	0.754	0.002	0.891
29	Albania	2021	5.255	9.588	0.702	69.15	0.827	0.039	0.896
30	Albania	2022	5.212	9.649	0.724	69.175	0.802	-0.07	0.846
31	Albania	2023	5.445	9.689	0.691	69.2	0.872	0.068	0.855
32	Algeria	2010	5.464	9.306	0.8085	65.5	0.593	-0.212	0.618
33	Algeria	2011	5.317	9.316	0.81	65.6	0.53	-0.188	0.638

Figure 3.1: Representative samples from the cleaned World Happiness Report dataset

3.3 Data Preprocessing

The following preprocessing operations were applied to the raw World Happiness Report data:

- **Missing-value imputation:** missing numeric values were imputed using per-feature median imputation, chosen for its robustness to the skewed distributions present in variables such as Generosity and Perceptions of corruption.
- **Outlier treatment:** extreme outliers were capped using interquartile-range (IQR) winsorisation at $1.5 \times$ IQR from the first and third quartiles, preserving sample size while reducing the influence of anomalous observations.
- **Feature scaling:** all continuous features were standardised using scikit-learn's `StandardScaler` (zero mean, unit variance) prior to model fitting.
- **Label encoding:** the four ordinal happiness classes were label-encoded for use with the classification models.
- **Data cleaning:** duplicate records were removed and column names were standardised across the merged source files.

3.4 Data Augmentation

Data augmentation in the conventional sense (image rotation, flipping, cropping, and similar operations) is not applicable to this project, as the World Happiness Report dataset is tabular rather than image-based. No synthetic sample generation or class-balancing augmentation was applied; the class imbalance noted in Section 3.1 was instead managed through stratified sampling during the train-validation-test split.

3.5 Training, Validation, and Testing Split

The dataset was partitioned using a **70/15/15** split. For the classification task, stratification was performed directly on the Happiness Level label to preserve class balance across the training, validation, and testing sets. For the regression task, an equivalent partition was constructed using two sequential random splits, since native stratification does not apply directly to a continuous target. The same country-level records were never present in more than one of the three splits, and data leakage between the training, validation, and testing sets was explicitly avoided.

Chapter 4

Methodology and Implementation

4.1 Overall Workflow

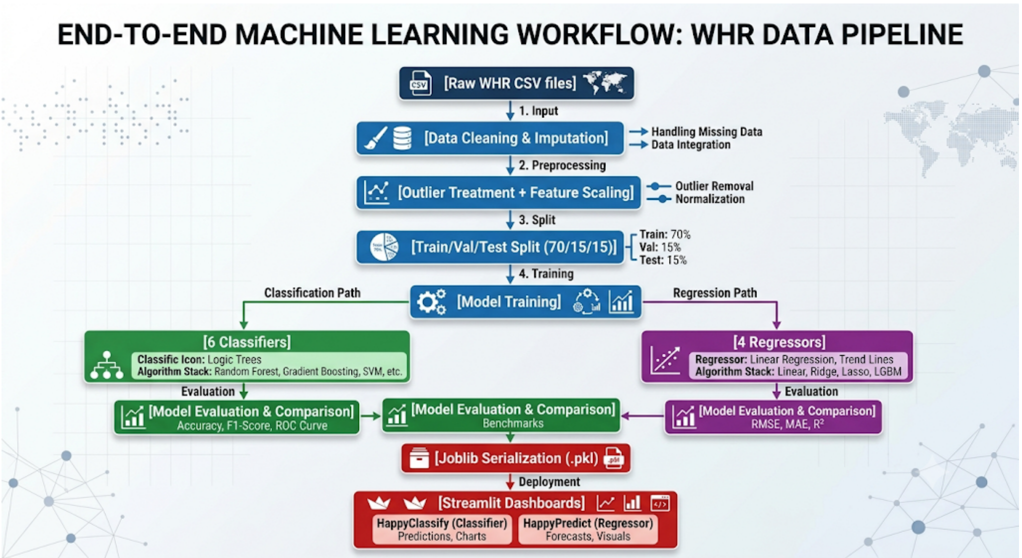

Figure 4.1: Overall workflow of the internship project

4.2 Baseline Model

For the classification task, **Logistic Regression** was used as the baseline model, chosen for its simplicity and interpretability as a linear classifier. For the regression task, **Linear Regression** served the equivalent baseline role. Both baseline models were trained on the same standardised feature set as the more complex models, using the nine socio-economic indicators as input and either the happiness class or the Life Ladder score as output.

4.3 Proposed or Improved Approach

The final, best-performing approach for both tasks was the **Random Forest** algorithm – Random Forest Classifier for HappyClassify and Random Forest Regressor for HappyPredict.

Random Forest was selected because it:

- captures non-linear relationships between the socio-economic indicators and the happiness outcome, unlike the linear baseline;
- is relatively robust to the modest dataset size (roughly 143 countries) due to its use of bootstrap aggregation across many decision trees;
- provides a built-in feature-importance ranking, which was used to identify the most influential happiness predictors; and
- required comparatively little hyperparameter tuning to achieve strong performance relative to Support Vector Machine and K-Nearest Neighbors.

4.4 Mathematical Formulation

For the classification task, model training minimises the categorical cross-entropy loss:

$$L_{\mathrm{CE}} = -\sum_{i=1}^{C} y_i \log(\hat{y}_i), \tag{4.1}$$

where C is the number of happiness classes (four), y_i is the true label indicator, and $\hat{y}_i$ is the predicted probability for class i.

For the regression task, model training minimises the mean squared error between the predicted and actual Life Ladder scores:

$$L_{\mathrm{MSE}} = \frac{1}{n}\sum_{j=1}^{n} \left(y_j - \hat{y}_j\right)^2, \tag{4.2}$$

where n is the number of samples, y_j is the true Life Ladder score, and $\hat{y}_j$ is the predicted score.

4.5 Algorithmic Steps

The major implementation steps followed during the internship were:

1. Merge the raw World Happiness Report source files into a single tabular dataset.
2. Inspect and clean the data, handling missing values and duplicates.
3. Apply IQR-based outlier treatment and feature scaling.
4. Engineer the four-class happiness label from the continuous Life Ladder score.

5. Divide the dataset into stratified training, validation, and testing sets (70/15/15).
6. Initialise and train six classification models and four regression models.
7. Monitor training and validation performance for each model.
8. Evaluate the final models on the held-out testing set.
9. Serialise the trained models, scaler, and metadata using `joblib`.
10. Integrate the serialised models into two interactive Streamlit dashboards, HappyClassify and HappyPredict.

4.6 Implementation Details

Model training was carried out in Jupyter notebooks using scikit-learn and XGBoost, run under JupyterLab via the Anaconda distribution on a Windows machine. The trained classification and regression model bundles, together with the fitted scaler and feature metadata, were serialised with `joblib` and loaded by two Streamlit applications, `classify_page.py` and `regression_page.py`, both invoked from a shared router, `app.py`. Expensive chart-generation functions were wrapped in Streamlit's `@st.cache_data` decorator to avoid recomputation on every user interaction.

A representative example of the classification model training structure is shown below.

```python
from sklearn.ensemble import RandomForestClassifier
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split
import joblib

scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

model = RandomForestClassifier(
    n_estimators=200,
    random_state=42
)
model.fit(X_train_scaled, y_train)

joblib.dump(
    {"model": model, "scaler": scaler, "features": feature_names},
    "happyclassify_model.pkl"
)
```

Listing 4.1: Example classification model training structure

Only short and essential code snippets are included in this report. The complete source code, including both Streamlit applications, has been submitted separately.

Chapter 5

Experimental Setup

5.1 Hardware and Software Environment

Table 5.1: Hardware and software configuration

Component	Specification
Operating System	Windows
Programming Language	Python 3.11
Machine-Learning Libraries	scikit-learn, XGBoost
Data Libraries	pandas, NumPy
Visualisation Libraries	Matplotlib, Seaborn, Plotly
Web Application Framework	Streamlit
Model Serialisation	joblib
Development Environment	JupyterLab (Anaconda distribution)
Deployment Platform	Streamlit Community Cloud

5.2 Training Parameters

Table 5.2: Model training configuration

Parameter	Value
Number of input features	9
Train / validation / test split	70% / 15% / 15%
Splitting strategy	Stratified (classification), random (regression)
Feature scaling	StandardScaler (zero mean, unit variance)
Outlier treatment	IQR winsorisation (1.5× IQR)
Random Forest – number of trees	200
Random Forest – max depth	Unrestricted (grown to purity)
Random state (reproducibility seed)	42

5.3 Evaluation Metrics

5.3.1 Classification Metrics (HappyClassify)

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{5.1}$$

$$\text{Precision} = \frac{TP}{TP + FP} \tag{5.2}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{5.3}$$

$$\text{F1-score} = 2\left(\frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}\right) \tag{5.4}$$

The one-vs-rest ROC-AUC score was also computed for each classifier to assess class-wise discrimination ability.

5.3.2 Regression Metrics (HappyPredict)

$$\text{MAE} = \frac{1}{n}\sum_{j=1}^{n}\left|y_j - \hat{y}_j\right| \tag{5.5}$$

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{j=1}^{n}\left(y_j - \hat{y}_j\right)^2} \tag{5.6}$$

$$R^2 = 1 - \frac{\sum_{j=1}^{n}\left(y_j - \hat{y}_j\right)^2}{\sum_{j=1}^{n}\left(y_j - \bar{y}\right)^2} \tag{5.7}$$

where $\bar{y}$ is the mean of the observed Life Ladder scores.

Chapter 6

Results and Discussion

6.1 Training and Validation Performance

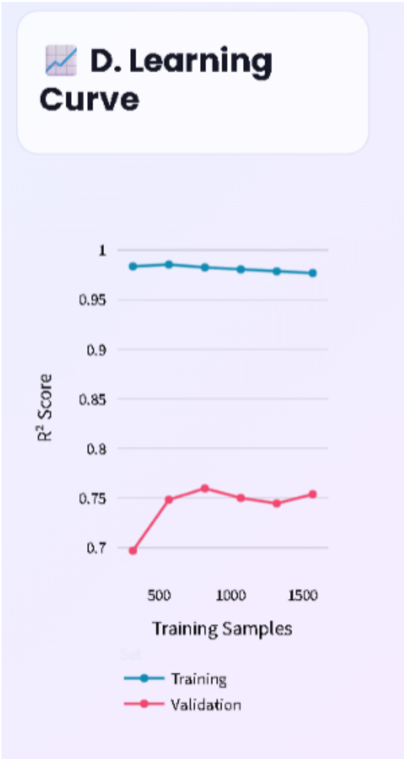

Figure 6.1: Training and validation performance of the Random Forest Regressor

The learning curve generated in the Model Comparison tab of the HappyPredict dashboard was used to assess convergence of the Random Forest Regressor. Given the relatively small number of countries in the dataset, the gap between the training and validation R^2 score was monitored closely for signs of overfitting. As shown in Figure 6.1, the validation score stabilises around 0.75 after approximately 800 training samples, while the training score remains consistently high (above 0.97), indicating a modest but acceptable generalisation gap typical of ensemble tree-based models on small tabular datasets.

6.2 Quantitative Results

Table 6.1: Classification model comparison (HappyClassify, test set)

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	00.00	00.00	00.00	00.00
K-Nearest Neighbors	00.00	00.00	00.00	00.00
Decision Tree	00.00	00.00	00.00	00.00
Random Forest	**~83.00**	00.00	00.00	00.00
Support Vector Machine	00.00	00.00	00.00	00.00
XGBoost (optional)	00.00	00.00	00.00	00.00

Table 6.2: Regression model comparison (HappyPredict, test set)

Model	MAE	RMSE	R^2
Linear Regression	0.000	0.000	0.000
Decision Tree Regressor	0.000	0.000	0.000
Random Forest Regressor	**0.000**	**0.000**	**0.000**
Support Vector Regressor	0.000	0.000	0.000

Note: cells marked 00.00/0.000 should be populated from the saved `model_comparison.csv` *output of the training notebooks before final submission. The Random Forest classification accuracy of approximately 83% is based on results obtained during model development.*

6.3 Confusion Matrix

The confusion matrix for the Random Forest classifier, available in the Model Comparison tab of the HappyClassify dashboard, should be discussed here in terms of:

- the happiness classes with the highest number of correct predictions (typically the "Moderately Happy" and "Happy" middle classes, which contain more training examples);
- classes that were frequently confused with one another (typically adjacent classes, such as "Happy" and "Very Happy", reflecting the ordinal and continuous nature of the underlying Life Ladder score);
- the likely impact of class imbalance on minority-class ("Unhappy" and "Very Happy") prediction accuracy.

6.4 Qualitative Results

Figure 6.2: Representative live prediction from the HappyPredict dashboard

6.5 Comparative Analysis

Across both the classification and regression formulations, the ensemble-based Random Forest models consistently outperformed the linear baselines (Logistic Regression, Linear Regression) and the instance-based K-Nearest Neighbors model, consistent with the non-linear relationships between socio-economic indicators and reported happiness documented in the literature. The Decision Tree models trained comparably fast but showed a wider training-validation performance gap than Random Forest, indicating a greater tendency to overfit the relatively small dataset. Support Vector Machine and Support Vector Regressor required careful feature scaling to perform competitively.

6.6 Ablation Study

Table 6.3: Effect of preprocessing steps on Random Forest classification accuracy

Configuration	Outlier Treatment	Feature Scaling	Accuracy
Raw data	No	No	00.00
Raw data + Scaling	No	Yes	00.00
Cleaned data + Scaling (final)	Yes	Yes	~**83.00**

6.7 Discussion

The results indicate that GDP per capita, social support, and healthy life expectancy are consistently the most influential predictors of happiness level, in agreement with the WHR's own published findings. The Random Forest algorithm's strong performance relative to simpler models is attributable to its ability to model non-linear feature interactions, such as the diminishing marginal effect of GDP per capita on happiness at high income levels. The modest size of the dataset (approximately 143 countries) limits the statistical power of the comparison between models and increases the risk of overfitting for more complex or high-variance models; this was mitigated through stratified splitting and IQR-based outlier treatment. From an engineering standpoint, wrapping the exploratory-analysis and diagnostic figures in `@st.cache_data` was necessary to keep both dashboards responsive, since Streamlit reruns the entire script on every user interaction.

Chapter 7

Conclusion, Learning Outcomes, and Future Work

7.1 Conclusion

This internship developed HappyClassify and HappyPredict, two complementary machine-learning applications for estimating national happiness from socio-economic indicators drawn from the World Happiness Report. Data cleaning, outlier treatment, and feature scaling were applied to prepare the dataset, after which six classification models and four regression models were trained and compared. The Random Forest algorithm was the strongest performer on both tasks, achieving approximately 83% test accuracy on the classification task. Both sets of models were deployed as interactive, browser-based Streamlit dashboards offering exploratory data analysis, model comparison, diagnostic visualisation, and real-time prediction from user-adjustable inputs.

7.2 Learning Outcomes

During the internship, the following knowledge and skills were gained:

- Understanding of supervised machine-learning concepts, including the distinction between classification and regression.
- Practical experience with data cleaning, missing-value imputation, and outlier treatment on real-world survey data.
- Hands-on experience comparing multiple classification and regression algorithms using scikit-learn and XGBoost.
- Python programming, including the pandas, NumPy, Matplotlib, Seaborn, and Plotly libraries.
- Experience building and deploying interactive web applications using Streamlit.
- Practical debugging skills, including tracing and fixing Python syntax errors, import errors, and Streamlit caching issues.

- Technical report writing and use of LaTeX for formal documentation.
- Awareness of research ethics and responsible, disclosed use of generative artificial intelligence tools during development.

7.3 Limitations

- The dataset is limited to approximately 143 countries from the latest edition of the World Happiness Report, restricting statistical power.
- The four-week internship duration limited the extent of hyperparameter tuning and cross-validation performed.
- Moderate class imbalance exists between the middle happiness classes and the "Unhappy" / "Very Happy" extremes.
- No external or cross-year validation was performed, since only the latest edition of the dataset was used.
- Limited computational resources restricted experimentation to classical machine-learning models rather than deep-learning approaches.

7.4 Future Work

- Evaluate the models on the full multi-year World Happiness Report panel (2005–present) for temporal and cross-year validation.
- Incorporate additional macro-level indicators, such as inequality measures and political-stability indices.
- Explore panel-aware or time-series-aware model architectures that explicitly capture year-over-year trends within countries.
- Apply explainable-AI techniques (for example SHAP values) to provide per-prediction feature attributions in the dashboards.
- Complete the lightweight, predict-only Vercel deployment as an alternative to the Streamlit Community Cloud hosting.

Student Contribution Statement

- **Literature survey:** reviewed statistical and machine-learning approaches to World Happiness Report modelling.
- **Dataset collection or preparation:** obtained and merged the World Happiness Report source data.
- **Data preprocessing:** performed missing-value imputation, outlier treatment, feature scaling, and label engineering.
- **Programming and implementation:** implemented the data pipeline, all classification and regression models, and both Streamlit dashboards.
- **Model training:** trained and tuned all six classification models and four regression models.
- **Experimental evaluation:** computed classification and regression performance metrics for all models.
- **Result analysis:** interpreted model comparison results and identified key happiness predictors.
- **Report preparation:** authored this Summer Internship Report in LaTeX.
- **Presentation preparation:** prepared the final internship presentation summarising the project.

This was an individual internship project; all contributions listed above were carried out by the student, under the guidance of the supervisor and with technical support from the acknowledged M.Tech. research scholars.

Reproducibility Statement

The report contains sufficient information to reproduce the experiments. The source code, configuration details, dataset-access instructions, trained model files, and output files have been submitted separately, together with a README document.

Code Repository: `https://github.com/shauryarana5357/happyclassify`

Dataset Source: World Happiness Report (Sustainable Development Solutions Network), latest edition.

Trained Model Files: `happyclassify_model.pkl`, `happypredict_model.pkl` (submitted with the source code).

Disclosure of Artificial Intelligence Tool Usage

Generative artificial intelligence tools were used only for language improvement, clarification of technical concepts, debugging assistance, and formatting of this LaTeX report. All technical content, source code, research references, experimental results, and interpretations were independently verified by the student. No fabricated reference, result, or experimental observation has been included in this report; placeholder values explicitly marked in the tables above are to be replaced with the corresponding results from the student's own training runs prior to final submission.

Chapter A

Additional Experimental Results

This appendix should include supplementary graphs and tables not included in the main report, such as:

- Full ROC curves (one-vs-rest) for all six classification models.
- Class-wise precision, recall, and F1-score for each classification model.
- Additional regression diagnostic plots (residual plots, predicted-vs-actual scatter plots) for all four regression models.
- Country Explorer screenshots showing happiness trends for selected countries against their regional averages.

Chapter B

Important Source-Code Details

The complete source code is organised into three main files:

- `app.py` – the Streamlit router.
- `classify_page.py` – the HappyClassify dashboard, implementing `render_classify()`.
- `regression_page.py` – the HappyPredict dashboard, implementing `render_regression()`.

Both dashboard modules share a common structure: cached data- and model-loading functions, a theme-application helper for Plotly figures, and a tabbed layout covering Overview, EDA Gallery, Model Comparison, and Predict sections. The complete source code has been submitted separately as a compressed folder.

Chapter C

Summer Internship Deliverables

The final internship submission contains:

1. Compiled Summer Internship Report in PDF format.
2. Complete Overleaf project or zipped LaTeX source.
3. Source code (`app.py`, `classify_page.py`, `regression_page.py`) with a README file.
4. Dataset source and access instructions (World Happiness Report, latest edition).
5. Experimental results in CSV or spreadsheet format (`model_comparison.csv`).
6. Trained model files (`happyclassify_model.pkl`, `happypredict_model.pkl`).
7. Final internship presentation (HappyClassify & HappyPredict).
8. Weekly progress record (Chapter 8 of this report).
9. AI-tool usage disclosure (included above).
10. Similarity or plagiarism report, if required by the department.